

Sarthak Parikh

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Introduction

MSc CS student at LMU Munich specializing in AI4Science, Interpretability, and the Mathematics of AI. During my MSc, I built machine learning models for scientific domains and analyzed their internal representations.

Education

Ludwig Maximilian University of Munich

April 2020-March 2024

BSc in Computer Science with a Minor in Statistics

- **Final Grade:** 1.7
- **Relevant Coursework:** Intro2ML, Linear Algebra, Analysis, Theory of Computation, Data Mining
- **Thesis:** Enhancing Data Valuation in Explainable AI: An STI-KNN and Reinforcement Learning Approach

Ludwig Maximilian University of Munich

April 2024-Now

MSc in Computer Science

- **Current Grade:** 1.4
- **Relevant Coursework:** GenAI and Visual Synthesis, ML, Mathematical Foundations of Machine Learning (I/II), Foundation Models, Algorithmic Bioinformatics (Graphs), Protein Prediction I/II, Large Model Alignment, ML with Knowledge Graphs, Selected Topics in QC (AlphaTensorQuantum), Formalization in Lean
- **Currently Studying:** Machine Learning and Structure, Information Geometry in ML, Causality

Experience

Research Assistant (Machine Learning for Binary Analysis)

Munich, Germany

Ludwig Maximilian University of Munich

Apr 2026 – Sep 2026

- Running two core research tracks in cross-modal representation learning, with two first-author manuscripts in preparation.
- **Representation Learning & Distillation:** Developing a Joint-Embedding Predictive Architecture (JEPA) for cross-modal mapping. Also distilled a 1.3B parameter LLM into a $6\times$ smaller encoder while maintaining state-of-the-art performance.
- **Mechanistic Interpretability:** Studying cross-modal attention mechanisms and latent alignments to extract human-understandable semantics and verify structural preservation.

Student Assistant

Munich, Germany

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Sep 2021-Sep 2025

- Performed data extraction, cleaning, pre-processing, and feature engineering on sensitive patient and research data, ensuring compliance with privacy regulations (GDPR)
- Created visualizations and reports using Matplotlib to communicate findings to researchers and medical staff.
- Managed Linux server infrastructure supporting data storage and analysis

Teaching Experience

Student Tutor

Munich, Germany

Ludwig Maximilian University of Munich

Oct 2024 – Mar 2026

- **Winter Semester 25/26:** Tutor for *Introduction to Informatics (InfoEinf)* and *FSV* (Formal Specification and Verification).
- **Winter Semester 24/25:** Tutor for *Introduction to Informatics (InfoEinf)*.

- Tasks include preparing and conducting weekly tutorials, reinforcing theoretical concepts, grading assignments and exams.

Projects

Generative Quantum Machine Learning (Quantum Computing Praktikum I & II)

- Implemented Restricted Boltzmann Machines (RBMs) utilizing Persistent Contrastive Divergence.
- Developed and evaluated Quantum Circuit Born Machines (QCBMs) for generative molecular modeling.

Structure-Preserving Circuit Partition for T-Count Optimization [↗](#)

- Proposed a novel graph-theoretic partitioning methodology using the METIS algorithm to address the scaling bottleneck of the AlphaTensor-Quantum compiler.
- Empirically validated the approach against a Monte Carlo baseline with PyZX, demonstrating up to 35.3% T-count reduction in modular arithmetic circuits by preserving local algebraic structures (gadgets).
- **Publication:** Featured in the Proceedings of the *Seminar on Selected Topics in Quantum Computing* (Top projects only)

Early Detection of Deceptive Intent via Activation Oracles [↗](#)

- Investigated whether sycophantic intent in Qwen-3-8B can be detected from internal activations before any meaningful text is generated.
- Used an Activation Oracle to classify hidden states as honest or deceptive across 200 TruthfulQA prompts; tokens 0–2 showed near-perfect latent separation ($p < 0.001$).
- MATS Application. (Borderline) Reject. Top 18%.

Formalizing Maximum Flow Algorithms in Lean [↗](#)

- Developed a formal implementation of the Ford-Fulkerson maximum flow algorithm utilizing the Lean 4 interactive theorem prover.

Generalization vs Memorization in Protein Language Models

- Used Embedding-based and Curvature-based Analysis (KFAC) to confirm non-trivial generalization

Activation Steering for Hallucination Reduction

- Engineered an activation steering approach for the LLaVA model to actively mitigate hallucinations in multi-turn QnA.
- Achieved superior performance metrics, successfully outperforming GPT-4V.

Protein Language Model Enhanced Pfam Classification

- Developed a lightweight architecture for Pfam classification, enhancing the ProtENN2 architecture by utilizing pre-trained ProtT5 language model embeddings

Evaluation of Decoupled Diffusion Transformers

- Probed the model representations to evaluate adequacy on downstream tasks like segmentation and classification

BA Thesis Project on Data Valuation, Grade: 1.0

- Developed a novel approach using Reinforcement Learning (A2C/PPO agents) within a custom environment to optimize sample weights for data valuation.
- Achieved an average model performance increase of $\sim 19\%$ per dataset through optimized sample weighting.

Technologies

Languages: Python, R, SQL, Haskell, C

ML & Scientific Libraries: PyTorch, PyTorch Geometric, Qiskit, Scikit-learn, NumPy, Pandas, NetworkX